Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Use of $v = \frac{2\pi r}{T}$ ecf **or** equivalent with $v = \omega r$ and $\omega = \frac{2\pi}{T}$ (1) (expect 65 000 m s^{-1}) Use of Doppler equation (1) Correct answer = 0.0934 nm (accept 1sf) (1) Note: ecf on previous values gives 0.187 nm, 0.264 nm and 0.066 nm (for full marks unless another clear mistake seen) Allow 2 marks max if d used instead of r (gives 0.187 nm) Allow 1 mark max if $\frac{1}{2}mv^2 = \frac{GMm}{r}$ (also gives 0.187 nm)	1 1	1		3	3	
	(c)		Centre of mass is nearer more massive star (1) Which will give a larger field OR but must be nearer smaller star for fields to match (1) Accept equivalent reverse argument		2		2	2	
	(d)	(i)	Force [due to both stars] are always equal [and opposite] Accept this is the point where the field is zero Accept masses of equal **mass and distance** either side		1		1		
		(ii)	Force due to black hole is 4×greater or 4.17×10^{30} [N] (1) Also force from other star (gives factor 5) or 1.04×10^{30} [N] (1)		2		2	1	